DHYANESH VISAKAN

Computer Science Engineering Student | Full-Stack Developer | AI/ML Enthusiast

Kattankulathur, Tamil Nadu, India | LinkedIn: linkedin.com/in/dhyanesh-visakan-534a02271 | GitHub: github.com/Codeminning

Live Project: edutime-pro.vercel.app | Portfolio: dhyanesh-visakan.dev (Editable)

EDUCATION

SRM Institute of Science and Technology (SRMIST) — Kattankulathur, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.15

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Machine Learning, Computer Vision, Computer Architecture

TECHNICAL SKILLS

Languages: Python, C, C#, C++, JavaScript, SQL

Frontend Development: HTML5, CSS3, React, Next.js, Vite, Tailwind CSS

Backend Development: FastAPI, Node.js, Express.js, REST APIs

Databases & BaaS: Supabase, MongoDB, MySQL

AI, ML & Computer Vision: TensorFlow, YOLOv8, OpenCV, NumPy, Pandas, LangChain, RAG (Retrieval-Augmented Generation)

Tools & Platforms: Git, GitHub, Docker, Streamlit, Vercel, Visual Studio Code

FEATURED ENGINEERING PROJECTS

Real-Time Pothole Detection and Municipal Budget Planning System [Python, YOLOv8, OpenCV, NumPy, Streamlit, Folium]

- Developed an intelligent road-monitoring computer vision pipeline detecting road surface potholes from image & video feeds.
- Integrated bounding-box visualization, confidence scoring, GPS logging, and interactive Folium geospatial mapping.
- Reported experimental metrics: mAP@0.5 of 0.80, mAP@0.5:0.95 of 0.50, and best F1-score of 0.75 at 0.442 confidence threshold.
- Generated zone-wise maintenance budget summaries and automated municipal repair cost reports.

EduTime Pro — AI Study Planner [React, Vite, Supabase, JavaScript, Vercel]

- Architected and deployed a production-ready student productivity web application (Live: https://edutime-pro.vercel.app/).
- Engineered academic scheduling, task organization algorithms, and persistent cloud storage using Supabase BaaS.

Enterprise Document Intelligence RAG Chatbot [FastAPI, React, LangChain, ChromaDB, Gemini API]

- Built a retrieval-augmented generation (RAG) system with document ingestion, text chunking, and ChromaDB vector search.
- Implemented grounded conversational question answering preventing hallucinations via context-guided prompt engineering.

WithLife — Smart Blood Bank Application [JavaScript, Supabase, Modern Web Development]

- Engineered donor information registries, real-time inventory tracking, and urgent blood request notification workflows.

CERTIFICATIONS & PRACTICAL WORKSHOPS

- Foundational C# with Microsoft — Microsoft & freeCodeCamp, 2025 (Reported Score: 91.3%)
- Embedded Systems Workshop — Top Engineers in association with IIT Madras (February 2025)
- C Programming Mastery Course — Udemy (November 2023, Completed 25.5-Hour Curriculum)
- Data Structures & Algorithms Course — Udemy (November 2024)
- Python Bronze Badge — HackerRank

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- National Cadet Corps (NCC) — Active cadet experience with verified leadership rank and drill discipline.
- Institution of Engineers (India) — Student member participating in technical seminars and engineering sessions.
- Rifle-Shooting — Competitive marksman experience demonstrating precision focus and sportsmanship under official record.
- Continuous Engineering Workshops — Actively participating in robotics, hardware-software interfacing, and AI bootcamps.